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Q1. Backpropagation multiplies the local derivatives of every layer. With sigmoid or tanh these are smaller than 1, so the gradient shrinks exponentially with depth and the first layers barely learn. Residual (skip) connections fix this by giving the gradient a direct path back; ReLU also helps since its derivative is 1 for positive inputs.

Q2. out = (W - K + 2P)/S + 1 = (32 - 5 + 2*1)/1 + 1 = 30. The output is 30 x 30.

Q3.

```
import math
def softmax(x):
    exps = [math.exp(v) for v in x]
    s = sum(exps)
    return [e / s for e in exps]
```